How predictive is diffusion MRI microstructure? A reproducible benchmark across datasets, features and models

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Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility. We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models. It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. This shows that standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI.

1 Introduction

Diffusion magnetic resonance imaging (dMRI) enables the in vivo representation of tissue microstructure and provides biologically meaningful biomarkers sensitive to axonal organization, myelination, and tissue complexity [19]. Leveraging these microstructural descriptors for predictive modeling has become increasingly common, with applications spanning demographic estimation [4, 15], clinical diagnosis [11, 29], and cognitive characterization [17]. The combination of diffusion-derived features and machine learning is therefore often viewed as a powerful framework for data-driven neurobiological inference.

While many report promising predictive performance, reported performance varies widely across studies, tasks, and datasets. Direct comparison between studies is often hard to achieve. Improvements are frequently attributed to model complexity or architectural innovation, yet it remains unclear to what extent these gains arise from representational choices, preprocessing variability, or evaluation design. As the number of dMRI–based prediction studies grows, the absence of a standardized and reproducible evaluation framework has become a critical bottleneck. Without controlled evaluation across datasets, preprocessing pipelines, and model families, it is difficult to assess methodological progress and understand the limitations of current approaches.

To address these limitations, we introduce an open-source diffusion MRI benchmark that enables controlled comparison across datasets, tissue types, microstructural representations, and model families. The benchmark explicitly quantifies performance variability across folds and configurations, isolates representation and tissue-specific effects, and evaluates the incremental value of diffusion-derived features beyond non-diffusion baselines. It provides the foundation for isolating microstructural signal from configuration-driven variability and for establishing comparable evaluation standards in diffusion MRI prediction.

2 Benchmark

Comparing predictive models across diffusion MRI studies remains challenging. Preprocessing pipelines differ substantially, microstructural information is represented in multiple ways (e.g., voxel-wise maps, ROI summaries, or learned embeddings), and evaluation protocols vary in data splits, cross-validation schemes, and hyperparameter selection. As a result, reported performance differences may reflect methodological inconsistencies rather than genuine modeling improvements. Limited code and configuration sharing further restrict reproducibility. Together, these factors make it difficult to determine which approaches generalize reliably and provide meaningful insights. To address these issues, we introduce a standardized benchmark that unifies preprocessing and feature extraction, evaluates both classical and deep learning models on regression and classification tasks, and spans multiple public datasets under a controlled evaluation protocol. We now describe the benchmark in detail whose overall structure is illustrated in Figure 1.

2.1 Datasets

To ensure reproducibility, we evaluate our benchmark on three publicly available datasets covering both healthy and clinical populations. **HCP**: The Human Connectome Project (HCP) Young Adult dataset [26] targets gender and age. It includes 1,036 healthy subjects after filtering (474 males, 562 females; ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b-values = 1000, 2000, 3000 s/mm^2, 90 directions/shell). **Cam-CAN**: The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset [22] targets gender and age. It includes 629 healthy subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b-values = 1000, 2000 s/mm^2, 30 directions each). **ABIDE II**:

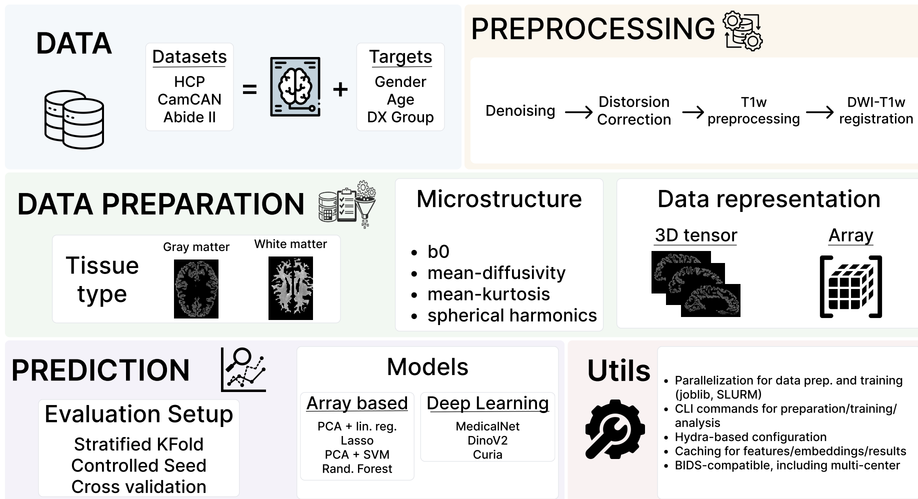

Fig. 1. Overview of the proposed diffusion MRI benchmark pipeline, illustrating the end-to-end process from preprocessing to microstructural feature extraction, predictive modeling, and evaluation.

The Autism Brain Imaging Data Exchange II (ABIDE II) dataset [10] targets Autism Spectrum Disorder (ASD) diagnosis. It includes 218 subjects (186 males, 32 females; ages 5-64), including 119 with ASD and 99 controls. As a multi-site dataset, ABIDE II data were acquired on various 3T scanners with highly heterogeneous, site-specific single-shell diffusion protocols. Across the included sites, spatial resolutions range from 0.94 to 3.0 mm, TR/TE from 5200-20244/78-101 ms, b-values include 1000, 1500, and 2500 s/mm^2, and diffusion directions vary between 32 and 64. This variability provides a realistic scenario for evaluating model robustness to scanner heterogeneity.

2.2 Data Preparation

Preprocessing Except for the HCP dataset, which has its own preprocessing pipeline, the DWI data were preprocessed using the diffusion-preprocessing package[1] which containerizes several other software dependencies such as dipy [12], sMRIPrep[2], FSL [24], ANTs [2], FreeSurfer [8] and AFNI [7]. The images were first denoised using the *Marchenko-Pastur* PCA method (MPPCA) [28, 6] implemented with dipy's *dipy_denoise_mppca* function. This denoising procedure identifies the principal components corresponding to noise and then removes them from the DWI data [28]. After MPPCA denoising, Gibbs ringing artifacts [27] were also removed using dipy's *dipy_gibbs_ringing* function. Next, to correct the distortions due to inhomogeneities of the magnetic field, FSL's

[1] https://github.com/man-shu/diffusion-preprocessing
[2] https://github.com/nipreps/smriprep

eddy correction [1] was used. Then this denoised and distortion-corrected DWI data were registered to the anatomical T1w image using FreeSurfer's *bbregister* function [13] with 12 degrees of freedom and boundary-based registration cost function. The anatomical T1w image was preprocessed using sMRIPrep.

Spatial normalization

Gray Matter Gray matter microstructure was represented on the subject-specific cortical surface reconstructed from T1-weighted images using the Desikan–Killiany atlas (aparc). For each subject, cortical meshes (white/pial surfaces) were generated and diffusion-derived scalar maps were computed in native diffusion space and rigidly aligned to the anatomical space. Voxelwise diffusion measures were then projected onto the individual cortical surface using ribbon-constrained sampling, preserving vertex-wise resolution (64k vertices per hemisphere) [13]. This subject-specific surface representation was chosen to respect cortical geometry, reduce partial-volume effects, and maintain high spatial specificity while enabling anatomically consistent regional correspondence via the atlas labels across subjects.

White Matter White matter microstructure was represented within a standardized tract space using the Tract-Based Spatial Statistics (TBSS) framework [23]. Diffusion-derived scalar maps were nonlinearly registered to standard space, projected onto the group-wise white matter skeleton to minimize partial volume effects and residual misalignment, and summarized using the JHU White-Matter Tractography Atlas [18]. Mean microstructural values were extracted from 48 major bundles defined in the atlas, yielding a compact tract-level representation per metric. We intentionally adopted this skeleton-based, atlas-constrained strategy rather than subject-specific tractography to enforce anatomical correspondence across subjects, reduce variability introduced by tracking parameters, and enhance robustness in a multi-dataset benchmarking setting. Although this representation sacrifices voxel-level granularity relative to the cortical surface analysis, it provides reproducible, homologous bundle definitions across datasets and mitigates registration and tract delineation inconsistencies—thereby strengthening cross-cohort comparability and methodological stability.

Microstructural representation To capture the biological properties of brain tissue, we compute a diverse set of microstructural metrics using established diffusion models. These metrics were selected to represent different facets of tissue architecture, ranging from basic diffusion magnitude to complex restricted environments:

- **Baseline ($b = 0$):** The averaged T2-weighted non-diffusion signal is included as a baseline to disentangle purely diffusion-driven predictive power from underlying structural T2 contrast.
- **Mean Diffusivity (MD):** Computed via Diffusion Tensor Imaging (DTI) [3], representing the overall magnitude of water diffusion.

– **Mean Kurtosis (MK):** Derived from Diffusion Kurtosis Imaging (DKI) [16], capturing the non-Gaussianity of diffusion as a proxy for microstructural heterogeneity and tissue complexity.
– **Spherical Harmonics (SH) Power:** A model-free representation obtained by computing the L2 norm of the SH coefficients [25] (up to order 6) directly from the diffusion signal, capturing signal complexity and anisotropy without assuming a specific tissue compartment model.

We apply a ventricular normalization strategy to avoid the effect of free water in the signal [21]. For each subject, voxel-wise diffusion metrics are divided by the mean metric value extracted from the cerebrospinal fluid (ventricles). Finally, extreme pathological values are clipped at the 99th percentile to ensure numerical stability during model training.

2.3 Prediction models

Our benchmark includes a wide spectrum of predictive models, ranging from linear baselines to advanced deep learning architectures. We first compare linear models (e.g., logistic regression, Lasso), Random Forests, and Support Vector Machines (SVM) [14]. Given the high dimensionality of the microstructural representations, we also evaluate these models coupled with Principal Component Analysis (PCA) for feature reduction.

We also benchmark several vision architectures. We leverage pre-trained models, specifically DINOv2 [20] and Curia [9], utilizing them as frozen feature extractors where representations are computed offline prior to training the prediction head. In contrast, the smaller 3D architecture MedicalNet [5] is fine-tuned on our diffusion datasets to adapt to the specific microstructural domains.

2.4 Benchmarks tasks and evaluation metrics

The selection of prediction tasks in our benchmark is driven by clinical relevance, data availability, and the need to align with established benchmarks in the neuroimaging literature. We evaluate models across two primary paradigms:

Classification For binary classification, we focus on tasks such as Autism Spectrum Disorder (ASD) diagnosis (ASD vs. Typically Developing) using the ABIDE II dataset, as well as gender prediction in the HCP and Cam-CAN cohorts. We report performance using balanced accuracy, taking into account class imbalances.

Regression We evaluate brain age estimation using the healthy cohorts from Cam-CAN. We report performance using R^2 and MAE.

2.5 Evaluation protocols

To rigorously evaluate the predictive models, we employ a stratified 5-fold cross-validation strategy. We also ensure fixed and reproducible splits through seed initialization. Strict separation between training and testing data is maintained at all stages; preprocessing steps, feature normalization, and hyperparameter tuning are fitted exclusively on the training set of each fold and subsequently applied to the held-out test set, preventing any information leakage.

3 Results

We evaluated the predictive power of diffusion MRI microstructural representations across datasets, tasks, tissue compartments, and model families. Rather than focusing only on peak performance, we analyze global patterns across configurations, allowing us to assess performance stability, tissue-specific contributions, and the added value of diffusion-derived features beyond non-diffusion baselines.

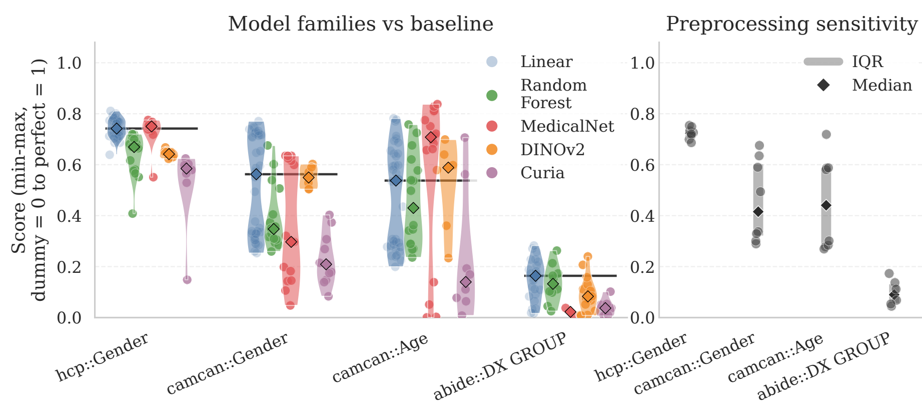

Fig. 2. Combined analysis of model-family performance and preprocessing sensitivity across tasks using a normalized score (0 = dummy, 1 = perfect) for cross-metric comparison. Left: Cross-validation results grouped by model family (points = models, violins = distributions, diamonds = medians; lines = linear baseline). Right: Sensitivity to preprocessing across feature × tissue configurations (dots = strategies; bars = IQR and range).

Figure 2 highlights two central findings. First, when evaluated on a common normalized scale, performance differences between model families are consistently modest. Regularized linear models remain competitive with deep learning architectures, independently of the task and data preparation strategy. Thus, the current data representation may not be sufficient to leverage the full potential of recent neural architectures. This likely reflects dataset size constraints; image-based models (Curia, DinoV2) were used as frozen feature extractors without fine-tuning, which typically demands larger cohorts to yield competitive performance.

Second, variability across preprocessing and microstructural parameterizations is substantial. In several tasks, the dispersion induced by feature and tissue configurations is larger than the differences observed between model families. This sensitivity indicates that predictive performance in diffusion MRI cannot be attributed to model choice alone. Instead, preprocessing strategy happens to be a major and often underreported source of variance. Conclusions drawn from a single representation or configuration may be unstable, therefore, underscoring the need for robustness analyses across preprocessing settings.

3.1 Impact of tissue type

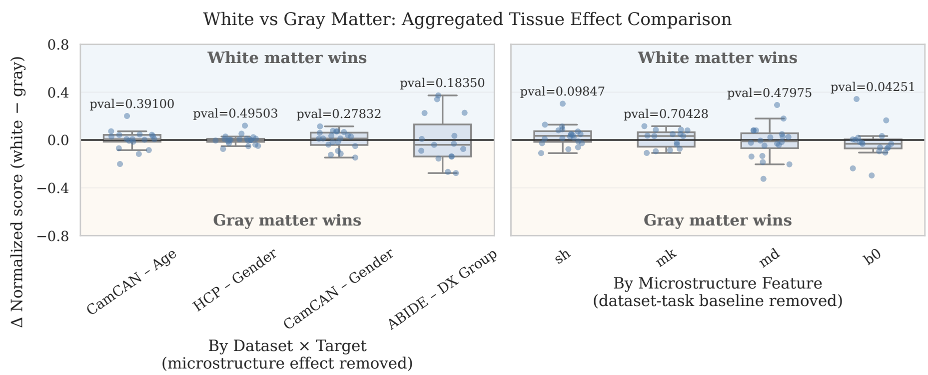

Fig. 3. Fold-level paired comparison of white vs. gray matter using normalized scores (0 = dummy, 1 = perfect). Each point shows the matched fold difference (Δ = white − gray) under identical settings. Left: tissue effects by dataset × target (microstructure baseline removed); Right: tissue effects by microstructure feature (dataset-task baseline removed). Boxplots summarize fold differences; p-values test mean effects against zero.

We next quantify the impact of tissue type by comparing the performance of models with white and gray matter under identical dataset–target–task–feature configurations. For each fold, performance is normalized relative to a dummy baseline and the difference Δ = white − gray is computed, giving a matched, configuration-specific tissue effect. To disentangle tissue influence from dataset and representation biases, we residualize the effects in two complementary ways: (i) removing feature-specific baselines to isolate dataset-dependent tissue effects, and (ii) removing dataset–task baselines to isolate feature-dependent tissue effects. Statistical inference is performed on configuration-level mean effects to account for fold dependence, testing whether the tissue difference distribution deviates from zero.

As shown in Figure 3, tissue effects are strongly dataset- and task-dependent. When stratified by dataset × target, gender classification in both HCP and CamCAN shows distributions centered near zero, indicating no systematic predominance of either tissue compartment for this task. In contrast, age prediction in CamCAN shows a slight tendency toward white-matter superiority, with a positive shift in the fold-difference distribution suggesting that white-matter microstructure carries marginally more age-relevant information. Diagnostic classification (DX_GROUP) in ABIDE shows the opposite pattern, with gray matter producing consistently higher performance, pointing to a gray-matter advantage for this clinical task.

When stratified by microstructural representation, most features do not show a consistent tissue preference, suggesting that the tissue effect is driven primarily by the task and dataset context rather than the choice of diffusion metric. Collectively, these findings demonstrate that tissue preference is neither universal nor negligible, but instead emerges from the interaction between dataset characteristics, prediction target, and microstructural representation.

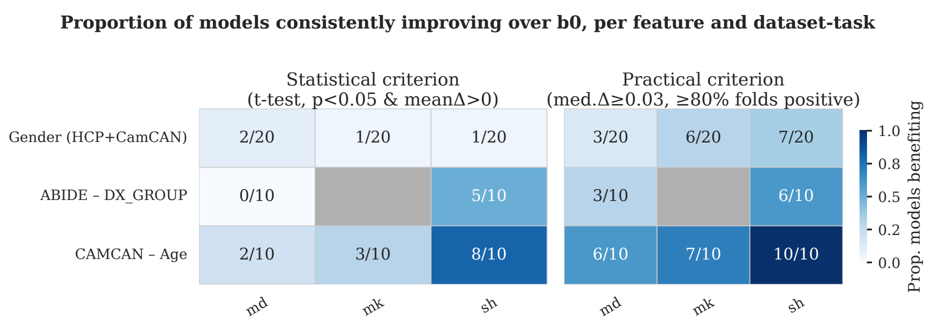

Fig. 4. Proportion of models improving over the non-diffusion baseline (b0), stratified by task and feature. Improvement is based on fold-level normalized differences (Δ = feature − b0). Left: statistical criterion (t-test against zero, $p < 0.05$ and mean $\Delta > 0$); Right: robustness criterion (median $\Delta \geq 0.03$ and $\geq$80% positive folds). Cells report the fraction of models meeting each criterion per target (rows) and feature (columns).

3.2 Impact of diffusion features

To assess whether diffusion-derived microstructural features provide predictive value beyond non-diffusion signal, we quantified model-wise improvements relative to b0. The results reveal that diffusion information does not uniformly enhance performance across tasks or datasets. For proxy tasks such as gender classification in HCP and CamCAN, diffusion features do not confer a systematic advantage over b0, indicating that non-diffusion signal (likely reflecting global anatomical properties such as brain size) already captures most of the discriminative information. In contrast, tasks more directly linked to microstructural variation show measurable gains from diffusion metrics. Age prediction in CamCAN and diagnostic classification (DX_GROUP) in ABIDE exhibit consistent improvements for several diffusion representations, particularly under linear models operating on array-based features (e.g., spherical harmonics).

Notably, deep learning models tend to exhibit positive gains in certain gender classification settings, likely reflecting their sensitivity to global morphological cues embedded in image-based representations rather than diffusion-specific contrast. Overall, these findings indicate that the added value of diffusion MRI depends strongly on the biological relevance of the task, the dataset characteristics, and the modeling framework, rather than representing a universal performance boost.

4 Conclusion

We introduced a diffusion MRI benchmark for demographics prediction in individuals across three datasets: Human Connectome Project, Cambridge Centre for Ageing and Neuroscience, and Autism Brain Imaging Data Exchange. Within a unified experimental framework, we systematically compared multiple microstructural representations (b0, MD, SH, MK) and model families spanning from linear to deep learning models. By controlling preprocessing, model selection, and evaluation procedures, the benchmark enables comparison across configurations and reduces methodological heterogeneity simplifying more transparent and reproducible comparisons in diffusion MRI prediction studies.

Our results suggest that predictive performance in diffusion MRI is highly configuration-dependent. In several settings, regularized linear models performed similarly to more complex approaches. More importantly, variability across folds, tissue types, and microstructural representations was often comparable to, or larger than, mean performance differences. Tissue type preferences were task-specific , and most microstructural features do not exhibit universal superiority. Diffusion-derived metrics provide measurable gains only in tasks closely aligned with microstructural variation, whereas for proxy tasks, non-diffusion signal can be equally informative.

This benchmark should not be interpreted as a definitive comparison of models or features, but rather as a controlled reference point. The included datasets, although widely used, remain moderate in size for modern machine learning, especially for deep learning approaches that typically require larger cohorts or fine-tuning. In addition, the current analysis focuses on within-dataset evaluation. The ability of models and representations to generalize across cohorts, scanners, and acquisition protocols remains an open question. This aspect was intentionally left outside the scope of the present work in order to maintain a clear focus on controlled within-dataset evaluation.

Several directions for future work follow naturally from these limitations. First, expanding the benchmark to include additional and larger datasets would improve statistical power and allow stronger conclusions, particularly for deep learning methods. Second, incorporating more clinically focused datasets with well-defined diagnostic outcomes could help clarify the clinical relevance of diffusion-derived features. Third, harmonization strategies across scanners and sites should be systematically evaluated, as variability in acquisition and preprocessing may substantially influence predictive performance. Finally, cross-cohort validation and domain adaptation experiments would help determine whether observed effects are stable under distributional shift.

Overall, we hope this benchmark contributes to ongoing efforts in the field toward more structured, reproducible, and variance-aware evaluation of diffusion MRI prediction models. By providing an open and controlled framework, this work aims to complement prior studies and support future investigations that build on shared standards.

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